

# Oluwatobi Melvyn Mayungbo

ML Engineer, MLOps and Production Systems

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## SUMMARY

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ML engineer who takes models from research to production. Converted a PyTorch model to ONNX Runtime for a 2.9× speedup and served it at 5.1 ms p99, deployed a model with Docker on AWS ECS Fargate with CI/CD, and writes tested, documented Python. B.Sc. in Statistics and teaches machine learning.

## SKILLS

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**Engineering and MLOps:** AWS ECS Fargate, Docker, CI/CD (GitHub Actions), FastAPI, Python, Pytest, SQL, DuckDB, PostgreSQL, Redis, GCP Cloud Run, Railway, Prefect, dbt-core, MLflow, Evidently, Git, Streamlit, React

**AI and LLM Systems:** ONNX Runtime, Retrieval-Augmented Generation (RAG), Hybrid Search (BM25, Dense Embeddings, RRF), LLM Evaluation and Guardrails, Prompt Engineering, Citation Verification, Model Context Protocol (MCP), Hugging Face

**Machine Learning:** PyTorch, Scikit-learn, Sequence Models (GRU, TCN, Transformers), LightGBM, XGBoost, Deep Learning, Model Calibration (Isotonic Regression, Temperature Scaling), SHAP, Feature Engineering

**Statistics and Causal Inference:** Statistical Inference, Experiment Design, Power Analysis, CUPED, Causal Inference (S/T/X-Learners, Causal Forests), Uplift Modeling, Qini and AUUC, Time-Series Backtesting, Logistic Regression, WOE Scorecards

**Data Analysis:** Pandas, NumPy, SQL, Anomaly Detection

**Payments and Risk:** Transaction Fraud Detection, Credit Risk Decisioning, Payment Reconciliation, Double-Entry Ledgers and Idempotency, Regulatory Compliance (CBN/NDPA), Demand Forecasting, Web3 Fraud Risk Scoring

## PROJECTS

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### Sequence-Based Transaction Fraud Detection System | [GitHub](#)

*Tech: Python, PyTorch, LightGBM, ONNX Runtime, FastAPI, Docker, SHAP*

- Deployed real-time scoring by exporting to ONNX Runtime (2.9× faster than eager PyTorch at batch size 1) and serving via FastAPI, reaching 5.1 ms p99 end-to-end latency against a 50 ms budget, with append-only per-decision audit logging and attention-based attribution of the prior transactions driving each flag.
- Built and compared three deep sequence architectures (GRU, Temporal Convolutional Network and a causally masked Transformer encoder) over customer transaction histories, with learned embeddings for high-cardinality merchant (~2.9k vocabulary), MCC and device-type categoricals; all models under 115k parameters and CPU-trainable in under 6 minutes.

### Telco Customer Churn, End to End on AWS | [GitHub](#)

*Tech: Python, XGBoost, MLflow, FastAPI, Gradio, Docker, GitHub Actions, AWS ECS Fargate, Application Load Balancer, CloudWatch*

- Served it through a FastAPI /predict endpoint with a Gradio UI, containerised with Docker and deployed on AWS ECS Fargate behind an Application Load Balancer, with CloudWatch logs and GitHub Actions CI/CD.
- Built a churn prediction model with feature engineering and an XGBoost classifier, with experiments tracked in MLflow.

### Fintech Compliance RAG Assistant | [GitHub](#) | [Live App](#)

*Tech: Python, Streamlit, BM25, ONNX Runtime, Reciprocal Rank Fusion, FastAPI, Docker, GCP Cloud Run*

- Deployed on GCP Cloud Run via Docker within a ~225MB peak memory footprint (2GB budget), with per-session and daily request caps in application code to bound API cost on a public endpoint.

### Self-Operating Demand Forecasting Platform | [GitHub](#)

*Tech: Python, Prefect, dbt-core, DuckDB, LightGBM, MLflow, Evidently, FastAPI, Docker, GitHub Actions*

- Built an end-to-end hourly demand forecasting system for NYC taxi zones, producing 24-hour quantile forecasts through an orchestrated Prefect pipeline spanning ingestion, dbt/DuckDB transformations, backtesting, drift monitoring and conditional retraining.

## EXPERIENCE

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**Machine Learning Instructor, SQI College of ICT**

March 2026 to Present

- Teach machine learning and deep learning to student cohorts, covering statistical foundations, classical ML algorithms, neural architectures, sequence models and production evaluation metrics.
- Instruct students on data analysis, feature engineering and model building with PyTorch, Scikit-Learn, Pandas and NumPy.

**Internal Auditor, NISER**

May 2024 to July 2024

- Analysed financial and operational records to identify inconsistencies and irregularities, performing manual anomaly detection across complex transactional datasets.

**EDUCATION**

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**B.Sc. Statistics, University of Ibadan**

January 2021 to March 2025

**Professional Diploma in Artificial Intelligence, SQI College of ICT**

July 2025 to Present

**CERTIFICATIONS**

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- Google Data Analytics Professional Certificate
- Google Advanced Data Analytics Professional Certificate
- Claude Code In Action